

Assignment 1

1. Consider the steady-state Boltzmann transport equation for an infinite, homogeneous, isotropic medium with a delta-source located at $x = 0$.

Plot the exact solution of the equation for

$$c = 0, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95.$$

- (a) For each value of c , plot separately the asymptotic component and the transient component of the steady-state solution.

In addition, for each value of c , plot the relative contribution of the transient and asymptotic components as a function of x :

$$\frac{\phi_{\text{tr}}(x)}{\phi(x)} \quad \text{and} \quad \frac{\phi_{\text{as}}(x)}{\phi(x)},$$

where

$$\phi(x) = \phi_{\text{as}}(x) + \phi_{\text{tr}}(x).$$

- (b) Present the classical diffusion solution for the same cases.
 - (c) Present the asymptotic diffusion solution for the same cases.
 - (d) For each value of c , plot the relative error as a function of x for each diffusion approximation, compared with the exact solution.
2. Write a numerical code for the steady-state diffusion equation with a delta-source located at $x = 0$, using either the classical diffusion approximation or the asymptotic diffusion approximation.

Points for consideration: How should a delta-function source be modeled in a numerical scheme?

Compare the converged numerical results with the corresponding analytical solution for each diffusion approximation.

3. For $1.02 < c < 2$, compute the critical half-thickness $a/2$ in planar geometry and the critical radius R_c in spherical geometry.

Perform the calculation using:

- (a) Exact transport approximation, using either $z_0(c)$ or $\ell_0(c)$.
- (b) Diffusion approximation with Marshak boundary conditions.
- (c) Diffusion approximation with Mark boundary conditions.
- (d) Asymptotic diffusion approximation using the exact $\ell_0(c)$ or $z_0(c)$.
- (e) Asymptotic diffusion approximation using $\ell_0(c)$ obtained from a modified Marshak-like boundary condition.

Plot $a/2$ and R_c as functions of c , and compare the results.

4. Write a numerical code for the steady-state diffusion equation in spherical coordinates and determine the critical radius by finding the radius for which the eigenvalue satisfies $k = 1$.

Perform the calculation using both the classical diffusion approximation and the asymptotic diffusion approximation.

Use the iterative algorithm for calculating k that appears in Bell & Glasstone, pp. 190–192.

Compare the numerically obtained critical radius with the analytical result for each approximation.

5. The following realistic cross sections are taken from:

Avneet Sood, R. Arthur Forster, and D. Kent Parsons, “Analytical Benchmark Test Set for Criticality Code Verification,” *Progress in Nuclear Energy*, Vol. 42, No. 1, pp. 55–106, 2003.

Material	ν	Σ_f	Σ_c	Σ_s	Σ_t	c
Pu-239	3.24	0.081600	0.019584	0.225216	0.32640	1.50
U-235	2.70	0.065280	0.013056	0.248064	0.32640	1.30
H ₂ O	0.0	0.0	0.032640	0.293760	0.32640	0.90
Fe	0.0	0.0	0.00046512	0.23209488	0.23256	0.9980
Na	0.0	0.0	0.0	0.086368032	0.086368032	1.00

Using the data in the table, calculate the bare critical mass of U-235 and Pu-239.

Assume

$$\rho_{\text{U}} = 19 \text{ g/cm}^3, \quad \rho_{\text{Pu}} = 15.7 \text{ g/cm}^3.$$